

Lake Wateree Water Watch Monitoring Run: August 22, 2025

Compliance Index: 95%

Lake Wateree Water Watch (WW) in collaboration with the University of South Carolina (USC) conducted a routine monitoring event on August 22, 2025. YSI sonde measurements were taken at each of the 20 stations (Figure 1), and USC researchers were responsible for operating the sondes, collecting samples, and recording all measurements. A chart of the Secchi disk¹ measurements is provided at the end of this document, and a nutrient analysis will be included in the 2025 Annual Report.

USC Boat: Randy Kelly (Boat Operator), Archana Venkatachari (USC Researcher),
Volunteer Boat: Dick Foote (Boat Operator), Clara Taft (USC Researcher)

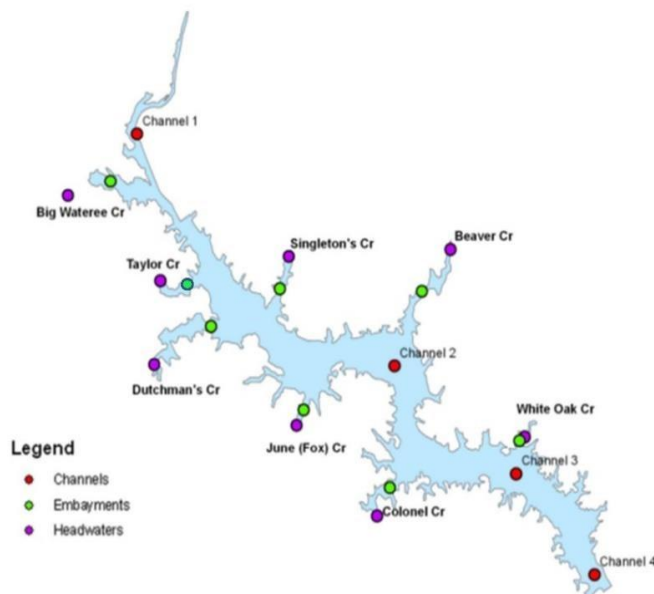


Table 1: Weather conditions at the Lake Wateree dam. Station ID: 388979 (Source: CLIMOD2)

	Sample Date	Prev. Day	Prev. 7-Day Avg.
Max. Temp.	89.0 (°F)	85.0	88.6
Min. Temp.	71.0 (°F)	68.0	70.9
Avg. Temp.	80 (°F)	71.0	79.7
Rainfall	0.05 (in.)	0.00	0.06
Temp. (°F)	Water	82.98	

Figure 1: Location of the twenty routine monitoring stations at Lake Wateree, South Carolina.

The compliance index for this month was 95%, and the historical compliance index for the same time of year (August) is 89.1%. The compliance index for this month indicates that of the 20 dissolved oxygen (mg/L) measurements, 19 of the 20 pH measurements, and 18 of the 20 turbidity (NTU) measurements taken at a 1-foot depth, all readings followed the South Carolina Department of Environmental Services (SCDES) water quality standards for freshwaters. The SCDES water quality criteria for dissolved oxygen (DO) states that no single sample should be below 4.0 mg/L and daily average samples should not be below 5 mg/L, and each of the 20 DO measurements

¹ Secchi disks are an alternate method of measuring turbidity that provides an estimate of water clarity.

recorded this month were within the criteria. 19 of the 20 of the pH measurements recorded at 1 foot were above the SCDES water quality criteria of between 6.0 and 8.5 for freshwaters, Channel 4 was slightly above the criteria (8.75). 2 of the 20 turbidity measurements recorded did not comply with the SCDES standard of 25 (NTU), both measurements were at Big Wateree Creek. The Secchi disk measurements for turbidity indicated an average surface water clarity of 0.59 meters for this monitoring event (Table 2). The average lake level on August 22nd, 2025, was 97.0 ft. as reported by the Duke Energy website, and the average water temperature was 82.98 °F (Table 1).

Table 2: All YSI at 1-ft and Secchi disc measurements by site name at Lake Wateree, South Carolina on August 22, 2025. Red numbers are out of SCDES compliance

Site Name	Date (MM/DD)	Time (HH:mn)	Cond μ S/cm	ODO mg/L	Turbidity FNU	pH	Temp $^{\circ}$ C	Secchi
Taylor Head	8/22/2025	9:34:10	92.9	4.36	8.58	6.27	29.147	0.5
Dutchman Head	8/22/2025	10:01:53	114.3	5.63	14.11	6.72	29.502	0.4
June Head	8/22/2025	10:38:45	105.4	4.79	5.34	6.97	28.494	0.65
Colonel Head	8/22/2025	11:05:06	114.2	6.47	12.64	7.37	30.041	0.4
WhiteOak Head	8/22/2025	11:37:25	102	7.94	7.6	7.49	27.335	0.5
Beaver Head	8/22/2025	12:11:31	89.1	7.07	13.24	7.16	26.742	0.4
Wateree Head	8/22/2025	12:56:35	87.9	8.02	32.74	7.84	27.485	0.3
Taylor Embay	8/22/2025	9:28:43	95.3	8.75	7.41	7.28	27.883	0.3
Wateree Embay	8/22/2025	9:46:35	83.9	7.02	25.46	6.84	28.485	0.4
Channel 1	8/22/2025	9:58:54	99.9	7.51	14.44	6.97	27.845	0.4
Dutchman Embay	8/22/2025	10:18:03	98	7.89	8.89	7.16	28.299	0.7
June Embay	8/22/2025	10:31:56	100.9	8.62	6.67	7.74	29.425	0.7
Channel 2	8/22/2025	10:45:55	99.3	8.97	7.46	8.04	28.972	0.7
Colonel Embay	8/22/2025	10:58:28	110.8	7.67	5.39	7.89	28.952	0.6
Channel 3	8/22/2025	11:11:39	102.5	8.61	5.85	8.27	28.473	0.7
Channel 4	8/22/2025	11:24:10	110.1	9.49	4.38	8.75	28.869	0.7
WhiteOak Embay	8/22/2025	11:40:02	108.4	8.86	5.06	8.48	28.03	0.7
Beaver Embay	8/22/2025	11:59:41	98.3	8.28	6.62	7.55	26.99	0.75
Singleton Head	8/22/2025	12:22:00	92.9	7.81	13.58	7.07	28.034	1.2
Singleton Embay	8/22/2025	12:27:28	94.9	8.49	10.06	7.39	27.43	0.75

A historical compliance index of 89.1% for August indicates that of the 1036 ambient water quality measurements taken at a 1-foot depth during monitoring events from previous years, 113 readings were not in compliance with the SCDES water quality standards. Of those readings, four DO measurements were below the water quality criteria threshold, 85 pH measurements were in exceedance of 8.5, and 24 turbidity measurements were above the SCDES standard of 25 (NTU). The compliance index for August 23, 2024, was 100% for 60 measurements recorded.